

Sophia Li

Sophomore | 770-778-1826 | sophiatl@stanford.edu

Profile

Motivated **Mechanical Engineering** and **Aeronautics & Astronautics** undergraduate at Stanford University with specialized experience in composite structural analysis, FEA, and full-cycle hardware design for high-performance vehicles. Proven track record in rapid prototyping, MATLAB-based data validation, and technical leadership on complex aerospace projects. Passionate about applying rigorous engineering analysis to drive innovation in flight.

Education

Mechanical Engineering, Stanford University

Palo Alto, CA | 09/2025 – 06/2029

- GPA: 4.048
- Relevant Coursework:
 - CS223A/ME320 – Introduction to Robotics
 - AA100 – Introduction to Aeronautics and Astronautics (propulsion fundamentals, rocket types)
 - CS106A, CS106B – Programming (Python and C++)
 - CME102 – Ordinary Differential Equations for Engineers
 - CME100 – Multivariable Calculus for Engineers
 - Completing thermodynamics coursework Fall 2026

Skills

- **CAD & FEA:** Autodesk Fusion 360, Abaqus (FEA), MATLAB (structural analysis, data visualization)
- **Aerodynamics:** XFLR5, airfoil analysis, lift distribution modeling
- **Hardware:** Rapid prototyping, mechanical design for flight
- **Manufacturing:** Wet layup composite fabrication, 3D printing, hand layup composites
- **Programming:** Python (CS106A, CS109), C++ (CS106B), MATLAB

Experience

Composite Wing Design, Analysis, & Test Engineer

Personal Project | 06/2026 – Present

- Built 2D/3D Abaqus composite shell models, validating structural response within 2% of analytical predictions.
- Designed and analyzed NACA 2412 wing; validated XFLR5 aerodynamic loading against published low-Reynolds data.
- Developed MATLAB scripts to process structural calculations, guiding laminate sizing and design decisions.
- Performing HyperX composite failure analysis using Tsai-Wu and Hashin criteria to evaluate design margins.
- Designed wet layup manufacturing plan; prototype fabrication in progress; 3-point bend test planned for experimental validation.

Mechanical Design Lead, Stanford DBF

Stanford University | 09/2025 – Present

- Lead mechanical design and prototyping for AIAA competition aircraft, managing 10 engineers across CAD, manufacturing, and testing phases.
- Designed and fabricated a two-stage servo-actuated banner release mechanism
- Managed full aircraft CAD integration in Fusion 360, coordinating fuselage, wing, and control surfaces
- Led design reviews and manufacturing timelines for rapid prototyping iteration.

Research Assistant, Stanford Freeman Spogli Institute (FSI)

Stanford University | 04/2026 – Present

- Facilitate high-level technical communication between faculty leads to accelerate the development of computer vision applications for clinical detection of Mild Cognitive Impairment (MCI).
- Authored and managed the Institutional Review Board (IRB) application, ensuring all research protocols met rigorous ethical and safety compliance standards for human-subject testing.
- Execute complex data processing and quantitative analysis of large-scale survey datasets regarding care robot technology adoption.
- Analyzed survey responses to identify trends in caregiver experiences and technology adoption.
- Developed data visualizations and summary statistics to communicate findings to research collaborators.

Volunteer Experience

Outreach, Stanford STEM Nights

Stanford University | 10/2025 – Present

- Orchestrated large-scale STEM engagement events, coordinating cross-functional teams to deliver technical demonstrations, ensuring rigorous safety compliance and effective public communication.
- Organizes social media.

Vice President, Outreach Director, Girls in STEM

Northview High School | 08/2022 – 05/2025

- Each month elementary school girls come and conduct science experiments.
- Organized a yearly collaboration with neighboring high schools to create a day-long camp with science experiments.
- Gives the opportunity for young girls in elementary school to experience fun science experiments, inspiring girls to pursue STEM as a career.

Awards

Academics

- 2024 + 2025 Morley Award Recipient (~23/1161)
 - Awarded to Georgia Tech's distance math students whose final grades for linear algebra and multivariable calculus were among the top 2% of the class.
- 2024 Yale Book Award Recipient (1/450)
 - One junior recognized annually at Northview High School for "outstanding personal character and intellectual promise." Chosen by counselors and teachers.
- 2024 National Chemistry Olympiad Qualifier (7th/~310)
 - Took a test on advanced chemistry topics (kinetics, equilibria, organic chemistry), and placed 7th/~310 students in the state of Georgia, qualifying to compete in the national exam.

Research

- 2024 Georgia Science and Engineering Fair – Top 3 Plant Sciences, 2nd honors (~50/226 Teams)
 - Presented detailed research on "Transgenerational Terminal Drought Stress Memory in Green Bean Germination (*Phaseolus Vulgaris*)."
- 2024 Fulton County Regional Science Fair – 1st place Plant Sciences